

Working with .resx Files Programmatically

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Content

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Work with .resx files programmatically

Summarize this article for me

Note

This article applies to .NET Framework. For information that applies to .NET 5+ (including .NET Core), see [Resources in .resx files](#).

Because XML resource (.resx) files must consist of well-defined XML, including a header that must follow a specific schema followed by data in name/value pairs, you may find that creating these files manually is error-prone. As an alternative, you can create .resx files programmatically by using types and members in the .NET Class Library. You can also use the .NET Class Library to retrieve resources that are stored in .resx files. This article explains how you can use the types and members in the [System.Resources](#) namespace to work with .resx files.

This article discusses working with XML (.resx) files that contain resources. For information on working with binary resource files that have been embedded in assemblies, see [ResourceManager](#).

Warning

There are also ways to work with .resx files other than programmatically. When you add a resource file to a [Visual Studio](#) project, Visual Studio provides an interface for creating and maintaining a .resx file, and automatically converts the .resx file to a .resources file at compile time. You can also use a text editor to manipulate a .resx file directly. However, to

avoid corrupting the file, be careful not to modify any binary information that is stored in the file.

Create a .resx file

You can use the [System.Resources.ResXResourceWriter](#) class to create a .resx file programmatically, by following these steps:

-

Instantiate a [ResXResourceWriter](#) object by calling the [ResXResourceWriter\(String\)](#) method and supplying the name of the .resx file. The file name must include the .resx extension. If you instantiate the [ResXResourceWriter](#) object in a `using` block, you do not explicitly have to call the [ResXResourceWriter.Close](#) method in step 3.

-

Call the [ResXResourceWriter.AddResource](#) method for each resource you want to add to the file. Use the overloads of this method to add string, object, and binary (byte array) data. If the resource is an object, it must be serializable.

-

Call the [ResXResourceWriter.Close](#) method to generate the resource file and to release all resources. If the [ResXResourceWriter](#) object was created within a `using` block, resources are written to the .resx file and the resources used by the [ResXResourceWriter](#) object are released at the end of the `using` block.

The resulting .resx file has the appropriate header and a `data` tag for each resource added by the [ResXResourceWriter.AddResource](#) method.

Warning

Do not use resource files to store passwords, security-sensitive information, or private data.

The following example creates a .resx file named CarResources.resx that stores six strings, an icon, and two application-defined objects (two `Automobile` objects). The `Automobile` class, which is defined and instantiated in the example, is tagged with the [SerializableAttribute](#) attribute.

```
using System;
using System.Drawing;
using System.Resources;

[Serializable()] public class Automobile
{
    private string carMake;
    private string carModel;
    private int carYear;
    private int carDoors;
    private int carCylinders;

    public Automobile(string make, string model, int year) :
        this(make, model, year, 0, 0)
    { }

    public Automobile(string make, string model, int year,
        int doors, int cylinders)
    {
        this.carMake = make;
        this.carModel = model;
        this.carYear = year;
        this.carDoors = doors;
        this.carCylinders = cylinders;
    }

    public string Make {
        get { return this.carMake; }
    }

    public string Model {
        get { return this.carModel; }
    }

    public int Year {
        get { return this.carYear; }
    }

    public int Doors {
        get { return this.carDoors; }
    }

    public int Cylinders {
        get { return this.carCylinders; }
    }
}
```

```

    }
}

public class Example
{
    public static void Main()
    {
        // Instantiate an Automobile object.
        Automobile car1 = new Automobile("Ford", "Model N", 1906, 0, 4);
        Automobile car2 = new Automobile("Ford", "Model T", 1909, 2, 4);
        // Define a resource file named CarResources.resx.
        using (ResXResourceWriter resx = new ResXResourceWriter(@".\CarResources.resx"))
        {
            resx.AddResource("Title", "Classic American Cars");
            resx.AddResource("HeaderString1", "Make");
            resx.AddResource("HeaderString2", "Model");
            resx.AddResource("HeaderString3", "Year");
            resx.AddResource("HeaderString4", "Doors");
            resx.AddResource("HeaderString5", "Cylinders");
            resx.AddResource("Information", SystemIcons.Information);
            resx.AddResource("EarlyAuto1", car1);
            resx.AddResource("EarlyAuto2", car2);
        }
    }
}

```

```
Imports System.Drawing
Imports System.Resources
```

```
<Serializable(>> Public Class Automobile
```

```
    Private carMake As String
```

```
    Private carModel As String
```

```
    Private carYear As Integer
```

```
    Private carDoors As Integer
```

```
    Private carCylinders As Integer
```

```
    Public Sub New(make As String, model As String, year As Integer)
```

```
        Me.New(make, model, year, 0, 0)
```

```
    End Sub
```

```
    Public Sub New(make As String, model As String, year As Integer,
        doors As Integer, cylinders As Integer)
```

```
        Me.carMake = make
```

```
        Me.carModel = model
```

```
        Me.carYear = year
```

```
        Me.carDoors = doors
```

```
        Me.carCylinders = cylinders
```

```
    End Sub
```

```
    Public ReadOnly Property Make As String
```

```
    Get
```

```
        Return Me.carMake
```

```
    End Get
```

```
End Property
```

```
    Public ReadOnly Property Model As String
```

```
    Get
```

```
        Return Me.carModel
```

```
    End Get
```

```
End Property
```

```
    Public ReadOnly Property Year As Integer
```

```
    Get
```

```
        Return Me.carYear
```

```
    End Get
```

```
End Property
```

```
    Public ReadOnly Property Doors As Integer
```

```
    Get
```

```
        Return Me.carDoors
```

```

        End Get
    End Property

    Public ReadOnly Property Cylinders As Integer
        Get
            Return Me.carCylinders
        End Get
    End Property
End Class

Module Example
    Public Sub Main()
        ' Instantiate an Automobile object.
        Dim car1 As New Automobile("Ford", "Model N", 1906, 0, 4)
        Dim car2 As New Automobile("Ford", "Model T", 1909, 2, 4)
        ' Define a resource file named CarResources.resx.
        Using resx As New ResXResourceWriter(".\CarResources.resx")
            resx.AddResource("Title", "Classic American Cars")
            resx.AddResource("HeaderString1", "Make")
            resx.AddResource("HeaderString2", "Model")
            resx.AddResource("HeaderString3", "Year")
            resx.AddResource("HeaderString4", "Doors")
            resx.AddResource("HeaderString5", "Cylinders")
            resx.AddResource("Information", SystemIcons.Information)
            resx.AddResource("EarlyAuto1", car1)
            resx.AddResource("EarlyAuto2", car2)
        End Using
    End Sub
End Module

```

Tip

You can also use [Visual Studio](#) to create .resx files. At compile time, Visual Studio uses the [Resource File Generator \(Resgen.exe\)](#) to convert the .resx file to a binary resource (.resources) file, and also embeds it in either an application assembly or a satellite assembly.

You cannot embed a .resx file in a runtime executable or compile it into a satellite assembly. You must convert your .resx file into a binary resource (.resources) file by using the [Resource File Generator \(Resgen.exe\)](#). The resulting .resources file can then be embedded in an application assembly or a satellite assembly. For more information, see [Create resource files](#).

Enumerate resources

In some cases, you may want to retrieve all resources, instead of a specific resource, from a .resx file. To do this, you can use the [System.Resources.ResXResourceReader](#) class, which provides an enumerator for all resources in the .resx file. The [System.Resources.ResXResourceReader](#) class implements [IDictionaryEnumerator](#), which returns a [DictionaryEntry](#) object that represents a particular resource for each iteration of the loop. Its [DictionaryEntry.Key](#) property returns the resource's key, and its [DictionaryEntry.Value](#) property returns the resource's value.

The following example creates a [ResXResourceReader](#) object for the CarResources.resx file created in the previous example and iterates through the resource file. It adds the two `Automobile` objects that are defined in the resource file to a [System.Collections.Generic.List<T>](#) object, and it adds five of the six strings to a [SortedList](#) object. The values in the [SortedList](#) object are converted to a parameter array, which is used to display column headings to the console. The `Automobile` property values are also displayed to the console.

```

using System;
using System.Collections;
using System.Collections.Generic;
using System.Resources;

public class Example
{
    public static void Main()
    {
        string resxFile = @".\CarResources.resx";
        List<Automobile> autos = new List<Automobile>();
        SortedList headers = new SortedList();

        using (ResXResourceReader resxReader = new ResXResourceReader(resxFile))
        {
            foreach (DictionaryEntry entry in resxReader) {
                if (((string) entry.Key).StartsWith("EarlyAuto"))
                    autos.Add((Automobile) entry.Value);
                else if (((string) entry.Key).StartsWith("Header"))
                    headers.Add((string) entry.Key, (string) entry.Value);
            }
        }

        string[] headerColumns = new string[headers.Count];
        headers.GetValueList().CopyTo(headerColumns, 0);
        Console.WriteLine("{0,-8} {1,-10} {2,-4}   {3,-5}   {4,-9}\n",
                           headerColumns);

        foreach (var auto in autos)
            Console.WriteLine("{0,-8} {1,-10} {2,4}   {3,5}   {4,9}",
                               auto.Make, auto.Model, auto.Year,
                               auto.Doors, auto.Cylinders);
    }
}

```

// The example displays the following output:

```

//      Make      Model      Year  Doors  Cylinders
//
//      Ford      Model N    1906      0        4
//      Ford      Model T    1909      2        4

```

```
Imports System.Collections
Imports System.Collections.Generic
Imports System.Resources

Module Example
    Public Sub Main()
        Dim resxFile As String = ".\CarResources.resx"
        Dim autos As New List(Of Automobile)
        Dim headers As New SortedList()

        Using resxReader As New ResXResourceReader(resxFile)
            For Each entry As DictionaryEntry In resxReader
                If CType(entry.Key, String).StartsWith("EarlyAuto") Then
                    autos.Add(CType(entry.Value, Automobile))
                Else If CType(entry.Key, String).StartsWith("Header") Then
                    headers.Add(CType(entry.Key, String), CType(entry.Value, String))
                End If
            Next
        End Using

        Dim headerColumns(headers.Count - 1) As String
        headers.GetValueList().CopyTo(headerColumns, 0)
        Console.WriteLine("{0,-8} {1,-10} {2,-4} {3,-5} {4,-9}",
                           headerColumns)

        Console.WriteLine()
        For Each auto In autos
            Console.WriteLine("{0,-8} {1,-10} {2,4} {3,5} {4,9}",
                               auto.Make, auto.Model, auto.Year,
                               auto.Doors, auto.Cylinders)
        Next
    End Sub
End Module
```

```
' The example displays the following output:
'
'      Make      Model      Year  Doors  Cylinders
'
'      Ford      Model N    1906      0        4
'      Ford      Model T    1909      2        4
```

Retrieve a specific resource

In addition to enumerating the items in a .resx file, you can retrieve a specific resource by name by using the [System.Resources.ResXResourceSet](#) class. The [ResourceSet.GetString\(String\)](#) method retrieves the value of a named string resource. The [ResourceSet.GetObject\(String\)](#) method retrieves the value of a named object or binary data. The method returns

an object that must then be cast (in C#) or converted (in Visual Basic) to an object of the appropriate type.

The following example retrieves a form's caption string and icon by their resource names. It also retrieves the application-defined `Automobile` objects used in the previous example and displays them in a `DataGridView` control.

```

using System;
using System.Collections.Generic;
using System.Drawing;
using System.Resources;
using System.Windows.Forms;

public class CarDisplayApp : Form
{
    private const string resxFile = @".\CarResources.resx";
    Automobile[] cars;

    public static void Main()
    {
        CarDisplayApp app = new CarDisplayApp();
        Application.Run(app);
    }

    public CarDisplayApp()
    {
        // Instantiate controls.
        PictureBox pictureBox = new PictureBox();
        pictureBox.Location = new Point(10, 10);
        this.Controls.Add(pictureBox);
        DataGridView grid = new DataGridView();
        grid.Location = new Point(10, 60);
        this.Controls.Add(grid);

        // Get resources from .resx file.
        using (ResXResourceSet resxSet = new ResXResourceSet(resxFile))
        {
            // Retrieve the string resource for the title.
            this.Text = resxSet.GetString("Title");
            // Retrieve the image.
            Icon image = (Icon) resxSet.GetObject("Information", true);
            if (image != null)
                pictureBox.Image = image.ToBitmap();

            // Retrieve Automobile objects.
            List<Automobile> carList = new List<Automobile>();
            string resName = "EarlyAuto";
            Automobile auto;
            int ctr = 1;
            do {
                auto = (Automobile) resxSet.GetObject(resName + ctr.ToString());
            }
        }
    }
}

```

```
        ctr++;  
        if (auto != null)  
            carList.Add(auto);  
    } while (auto != null);  
    cars = carList.ToArray();  
    grid.DataSource = cars;  
}  
}  
}
```

```
Imports System.Collections.Generic
Imports System.Drawing
Imports System.Resources
Imports System.Windows.Forms
```

```
Public Class CarDisplayApp : Inherits Form
```

```
    Private Const resxFile As String = ".\CarResources.resx"
    Dim cars() As Automobile
```

```
Public Shared Sub Main()
```

```
    Dim app As New CarDisplayApp()
    Application.Run(app)
End Sub
```

```
Public Sub New()
```

```
    ' Instantiate controls.
```

```
    Dim pictureBox As New PictureBox()
    pictureBox.Location = New Point(10, 10)
    Me.Controls.Add(pictureBox)
    Dim grid As New DataGridView()
    grid.Location = New Point(10, 60)
    Me.Controls.Add(grid)
```

```
    ' Get resources from .resx file.
```

```
    Using resxSet As New ResXResourceSet(resxFile)
```

```
        ' Retrieve the string resource for the title.
```

```
        Me.Text = resxSet.GetString("Title")
```

```
        ' Retrieve the image.
```

```
        Dim image As Icon = CType(resxSet.GetObject("Information", True), Icon)
```

```
        If image IsNot Nothing Then
```

```
            pictureBox.Image = image.ToBitmap()
```

```
        End If
```

```
        ' Retrieve Automobile objects.
```

```
        Dim carList As New List(Of Automobile)
```

```
        Dim resName As String = "EarlyAuto"
```

```
        Dim auto As Automobile
```

```
        Dim ctr As Integer = 1
```

```
        Do
```

```
            auto = CType(resxSet.GetObject(resName + ctr.ToString()), Automobile)
```

```
            ctr += 1
```

```
            If auto IsNot Nothing Then carList.Add(auto)
```

```
        Loop While auto IsNot Nothing
```

```
        cars = carList.ToArray()
```

```
        grid.DataSource = cars
    End Using
End Sub
End Class
```

Convert .resx files to binary .resources files

Converting .resx files to embedded binary resource (.resources) files has significant advantages. Although .resx files are easy to read and maintain during application development, they are rarely included with finished applications. If they are distributed with an application, they exist as separate files apart from the application executable and its accompanying libraries. In contrast, .resources files are embedded in the application executable or its accompanying assemblies. In addition, for localized applications, relying on .resx files at runtime places the responsibility for handling resource fallback on the developer. In contrast, if a set of satellite assemblies that contain embedded .resources files has been created, the common language runtime handles the resource fallback process.

To convert a .resx file to a .resources file, you use [Resource File Generator \(*resgen.exe*\)](#), which has the following basic syntax:

```
resgen.exe .resxFilename
```

The result is a binary resource file that has the same root file name as the .resx file and a .resources file extension. This file can then be compiled into an executable or a library at compile time. If you are using the Visual Basic compiler, use the following syntax to embed a .resources file in an application's executable:

```
vbc filename .vb -resource: .resourcesFilename
```

If you are using C#, the syntax is as follows:

```
csc filename .cs -resource: .resourcesFilename
```

The .resources file can also be embedded in a satellite assembly by using [Assembly Linker \(*al.exe*\)](#), which has the following basic syntax:

```
al resourcesFilename -out: assemblyFilename
```

See also

- [Create resource files](#)

- [Resource File Generator \(*resgen.exe*\)](#)
- [Assembly Linker \(*al.exe*\)](#)

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